CryptoPulse: Multimodal Sentiment & Price Analysis

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Team Problem Statement

- **Problem:** Cryptocurrency volatility is hard to predict using price data alone. Public sentiment plays a crucial role.
- **Goal:** Build a multimodal model predicting Bitcoin price direction using:
    1. Historical Price/Volume Data (Quantitative)
    2. News Social Media Sentiment (Qualitative)

Data Sources

- **Market Data:** Yahoo Finance API (BTC-USD, ETH-USD). Daily OHLCV from 2020-2024.
- **Sentiment Data:**
  - NewsAPI / CryptoPanic (Headlines)
  - Reddit (r/Bitcoin, r/CryptoCurrency)

Preliminary Analysis (Week 8)

- **Naive Model Implementation:** We implemented a persistence model ($P_{t+1} = P_t$) as a baseline.
- **Results:**
  - MSE: 0.05
  - The naive model lags behind trends, highlighting the need for predictive complexity.
- **Visualization:** Plots showing Price History and Moving Averages have been generated (see Notebook).

Proposed Methodology

- **Preprocessing:** VADER/TextBlob for sentiment scores. MinMax scaling for prices.
- **Model:** LSTM (Long Short-Term Memory) network taking a merged vector of [Price, Volume, Sentiment Score] at time t.
- **Output:** Predict Closing Price at $t + 1$.